Module 1:

INTRODUCTION

Computer plays an important role in our daily life. Anything we want we can get only in one mouse click. Speed, reliability and accuracy of the computer make it a powerful tool for different purposes. A very important and basic need of today's modern business world is the quick availability and processing of information using computer. One can easily get the type of required information within a fraction of a second. The project that I have taken is also in this category which is used in our daily life whenever we want to purchase some items we can easily get them at our home.

1.1 Objective:

The objective of project on Online Shopping Portal is to developing a GUI based automated system, which will cover all the information Related to the all products which is used in our daily life. For example – Mobiles Phones, Laptops, Clothes, Books, Electronic Items and many more. So by this GUI based automated system a user want to purchase something then it only a mouse click away to purchase these products.

1.2 Need of ONLINE SHOPPING PORTAL

The "**ONLINE SHOPPING PORTAL**" is developed according the current need in different Fields. This is online shopping Website which provides facility for purchasing Mobiles, Laptops, Camera and many more items. So by using this Online Shopping Portal users which want to purchase some products will first Register an account on this portal then Login through their Username and Password, and then Select items which they want to purchase and add them to cart and finally checkout by giving payment details. So by using this portal users can easily purchase products from their home.

1.3 PROFILE OF THE PROBLEM

One must know what the problem is before it can be solved. The basis for the online shopping portal is to buy products online and save the timing.

A Online shopping portal, who want to buy any product of their need, has to contact different Shoppers, before deciding upon a particular Product that best suit his needs, requirements and satisfaction. Moreover, most of the work involved in this development process has to be done manually which is very time consuming and cumbersome and also, it reduces the efficiency, accuracy.

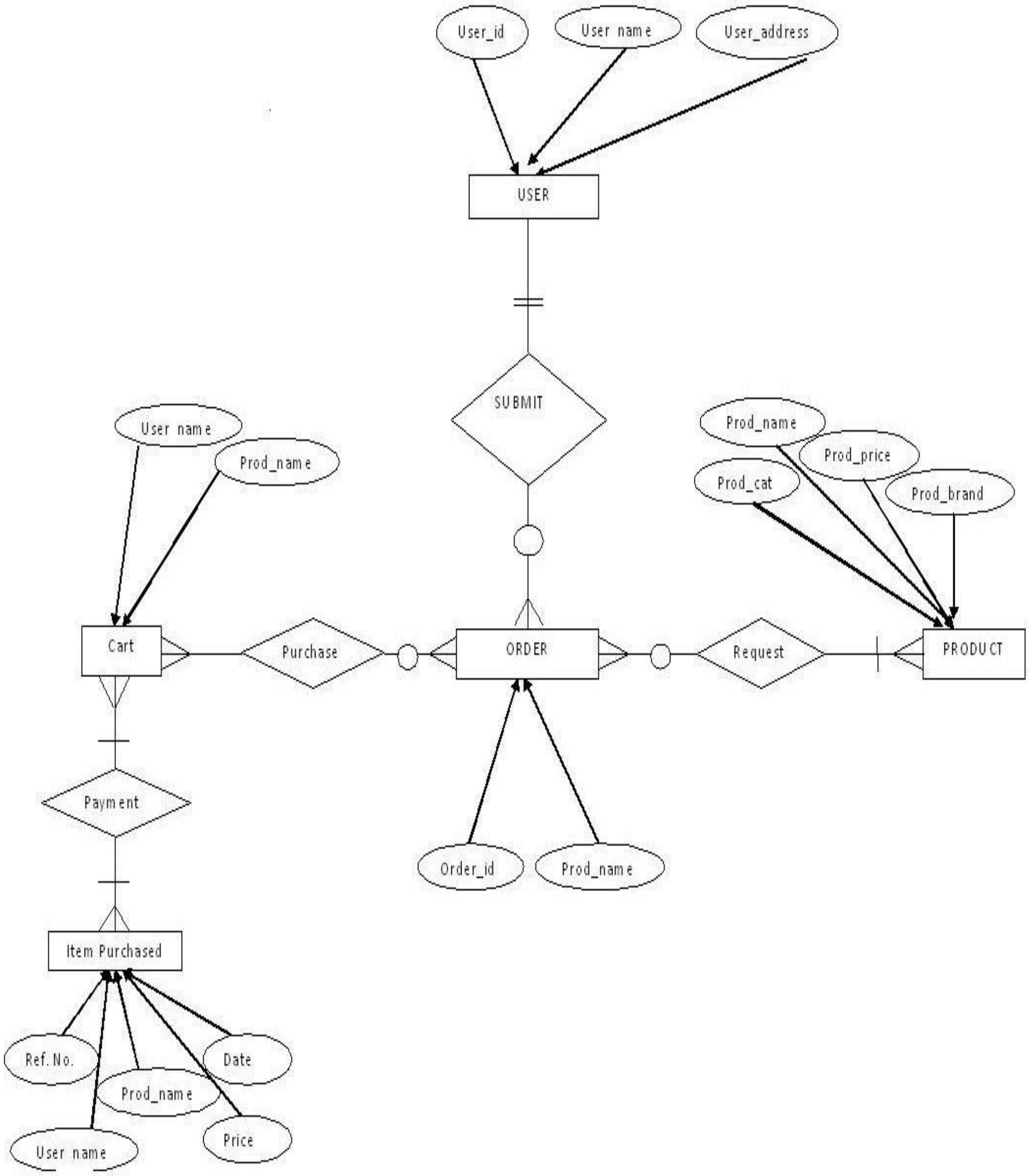
To know the facts and understanding of the problem in detail, ***System Analysis*** is carried out. It is the process of studying the business processes and procedures, generally referred to as business systems, to see how they can operate and whether improvement is needed.

Module 2:

PROJECT DESIGN

2.1 E R DIAGRAM

Illustrates the entities and relationships in the system, such as user, product, cart, order .
This diagram can help understand the data structure and guide database design and implementation.



2.2 **SCHEMA DIAGRAM**

Schema diagram formulates all the constrains that are to be applied on the data. A database schema defines its entities and the relationship among them. It contains a descriptive detail of the database, which can be depicted by means of schema diagrams

Cart

User name	Prod_name

Users

User_id	User_namr	User_address

Order

Order_id	Prod_name

Item Purchased

Ref.No	User_name	Prod_name	Price	Date

Product

Prod_name	Prod_cat	Prod_price	Prod_brand

Module 3:

HARDWARE & SOFTWARE REQUIRMENTS

3.1 At Developer Side

During system development, i have to design both static and dynamic website interfaces, create website functions and a database system, edit photos and pictures, so its has a set of software and hardware requirements.

Hardware Used

- Intel Dual Core Processor
- 160 GB Hard Disk Drive.
- 1GB RAM.
- O.S. – Windows XP SP2

Software Used

- WAMP SERVER
- MYSQL Database
- NOTEPAD
- MS PAINT

3.2 At System Users Side

The following is the requirements for the system users including members and administrators.

Hardware Requirements

- Intel Pentium 4 Processor
- 20 GB Hard Disk Drive.
- 256MB RAM.
- O.S. – Windows XP

Software Requirements

* Browser (IE 7.0 or Above, Mozilla Firefox, Google Chrome
* Browser Must be JavaScript Enabled

Module 4:
DESCRIPTION OF TOOLS AND TECHNOLOGIES

4.1Front End Details

Front End tool is used for give a Graphical user interface to system. By this we can make a system user friendly and more capable. I have chosen PHP as front end tool. Because it is an Open Source Technology, freely available and more familiar with any type of database.

XAMPP Server:

XAMPP stands for Cross-Platform (X), Apache (A), Maria DB (M), PHP (P) and Perl (P). Since XAMPP is simple, lightweight Apache distribution it is extremely easy for developers to create a local web server for testing and deployment purposes. Everything you needed is to set up a web server – server application (Apache), database (Maria DB), and scripting language (PHP). XAMPP works equally well on Linux, Mac, and Windows.

PHP:

PHP: Hypertext Preprocessor is a widely used, general-purpose scripting language that was originally designed for web development to produce dynamic web pages. For this purpose, PHP code is embedded into the HTML source document and interpreted by a web server with a PHP processor module, which generates the web page document. As a general-purpose programming language, PHP code is processed by an interpreter application in command-line mode performing desired operating system operations and producing program output on its standard output channel. It may also function as a graphical application. PHP is available as a processor for most modern web servers and as standalone interpreter on most operating systems and computing platforms.

PHP stores whole numbers in a platform-dependent range. This range is typically that of 32-bit signed integers. Unsigned integers are converted to signed values in certain situations; this behavior is different from other programming languages. Integer variables can be assigned using decimal (positive and negative), octal, and hexadecimal notations. Point numbers are also stored in a platform-specific range. They can be specified using floating point notation, or two forms of scientific notation. PHP has a native Boolean type that is similar to the native Boolean types in Java and C++. Using the Boolean type conversion rules, non-zero values are interpreted as true and zero as false, as in Perl and C++. The null data type represents a variable that has no value. The only value in the null data type is NULL. Variables of the "resource" type represent references to resources from external sources. These are typically created by functions from a particular extension, and can only be processed by functions from the same extension; examples include file, image, and database resources. Arrays can contain elements of any type that PHP can handle, including resources, objects, and even other arrays. Order is preserved in lists of values and in hashes with both keys and values, and the two can be intermingled. PHP also supports strings, which can be used with single quotes, double quotes, or heredoc syntax.

4.2 Back End Details

Back end part of a system is more important because it controls all the internal process of a system. I have choose oracle database as back end. Because it is word's Most Capable relational database and provide more security than others.

MySQL:

MySQL is the world's most popular open source database software, with over 100 million copies of its software downloaded or distributed throughout its history. With its superior speed, reliability, and ease of use, MySQL has become the preferred choice for Web, Web 2.0, SaaS, ISV, Telecom companies and forward-thinking corporate IT Managers because it eliminates the major problems associated with downtime, maintenance and administration for modern, online applications.

Many of the world's largest and fastest-growing organizations use MySQL to save time and money powering their high-volume Web sites, critical business systems, and packaged software — including industry leaders such as Yahoo!, Alcatel-Lucent, Google, Nokia, YouTube, Wikipedia, and Booking.com.

The flagship MySQL offering is MySQL Enterprise, a comprehensive set of productiontested software, proactive monitoring tools, and premium support services available in an affordable annual subscription.

MySQL is a key part of WAMP (Window, Apache, MySQL, PHP), the fast-growing open source enterprise software stack. More and more companies are using WAMP as an alternative to expensive proprietary software stacks because of its lower cost and freedom from platform lock-in.

Module 5:

IMPLEMENTATION

Implementation is the stage in the project where the theoretical design is turned into the working system and is giving confidence to the new system for the users i.e. will work efficiently and effectively. It involves careful planning, investigation of the current system and its constraints on implementation, design of method to achieve the change over, an evaluation, of change over methods. A part from planning major task of preparing the implementation is education of users. The more complex system is implemented, the more involved will be the system analysis and design effort required just for implementation. An implementation coordinating committee based on policies of individual organization has been appointed. The implementation process begins with preparing a plan for the implementation for the system. According to this plan, the activities are to be carried out; discussions may regarding the equipment have to be acquired to implement the new system.

Implementation is the final and important phase. The most critical stage is in achieving a successful new system and in giving the users confidence that the new system will work and be effective. The system can be implemented only after thorough testing is done and if it found to working according to the specification. This method also offers the greatest security since the old system can take over if the errors are found or inability to handle certain types of transaction while using the new system.

At the beginning of the development phase a preliminary implementation plan is created to schedule and manage the many different activities that must be integrated into plan. The implementation plan is updated throughout the Development phase, culminating in a changeover plan for the operation phase. The major elements of implementation plan are test plan, training plan, equipment installation plan, and a conversion plan.

There are three types of implementation:

Implementation of a computer system to replace a manual system.

Implementation of a new computer system to replace an existing system.

Implementation of a modified application to replace an existing one, using the same computer.

Successful implementation may not guarantee improvement in the organization using the new system, but improper installation will prevent it. It has been observed that even the best system cannot show good result if the analysts managing the implementation do not attend to every important detail. This is an area where the systems analysts need to work with utmost care.

5.1 Conversion Methods

A conversion is the process of changing from the old system to the new one. It must be properly planned and executed. Four methods are common in use. They are Parallel Systems, Direct Conversion, Pilot System and Phase In method.

Parallel systems:

The most secure method of converting from an old to new system is to run both systems in parallel. This method is safest one because it ensures that in case of any problem in using new system, the organization can still fall back to the old system without the loss of time and money.

5.2 STRUCTURE OF PROJECT

- Before Login
  - Login
  - Register
  - Forget Password
  - Administrator Login
  - About Us
  - Contact Us

- After Administrator Login
  - Edit Website Details
  - Add Brands
  - Add Category
  - Add Items
  - Delete Brands
  - Delete Category
  - Delete Items
  - Manage User
    - See Users
    - Users Shopping
    - Add Users
    - Delete Users
  - Logout

- After User Login
  - My Profile
    - Edit Profile
    - Change Password ➢ Buy Products
    - Categories (Controlled by Admin. Which can be add it dynamically according to their needs)
  - My Cart
  - My Shopping’s
  - Checkout
  - Logout

Module 6:

TEST CASES

- **System is properly linked or not** - Whether they are redirected to desired page or not.
- **Information passed** – If a page passes some parameter to another page then it should be checked that the page get the correct information, whatever is passed by the previous page.
- **Output should be correct** – Every functionality of the system should be checked properly whether it gives the right result or not generally test is performed with known results. If the output of the system is matched with that result the system is working fine.

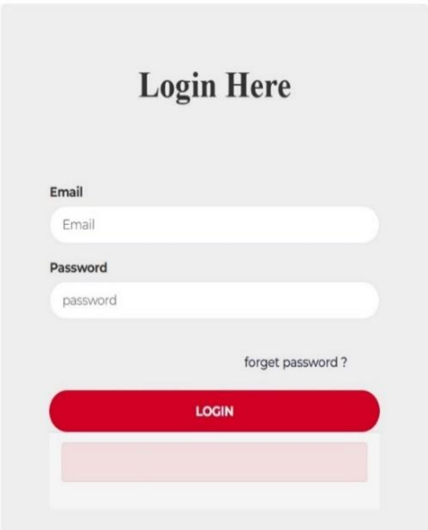
6.1 LOGIN FOR USER

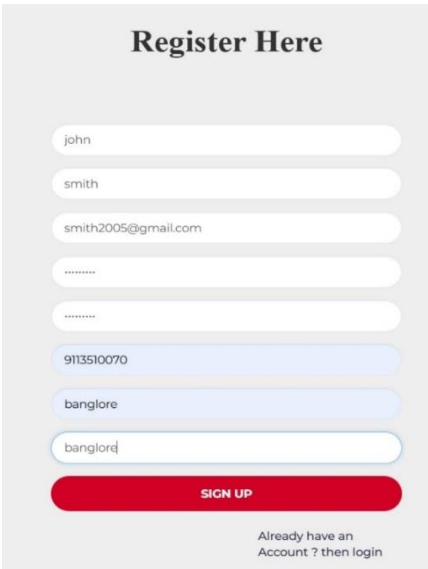
Serial No	Description	Expected Result	Actual Result	Result
1.	This page contains 2 fields user name and password and a login button to submit the information. User is entering correct information.	User home page should open after successful login.	Respective user home page is opening after successful login by user.	Passed
2.	If either user name or password is filled incorrect or left blank.	An error message should be displayed and user should be asked fill the information again.	When wrong information is entered by user then an error message is displayed.	Passed

6.2 USER REGISTRATION PAGE

Serial No	Description	Expected Result	Actual Result	Result
1.	User registration page 1 consist of detail information about User and a submit button to submit the information .Here user is entering correct information.	After submitting information User registration page 2 should be displayed.	After submitting information User registration page 2 is displayed.	Passed
2.	If the information entered by user in incorrect or left somewhere blank.	An error message should be displayed and ask the user to fill the information again.	An error message is occurred if the information is incorrect or left blank.	Passed

6.3 Test Result





Module 7:

RESULTS

7.1 Home page

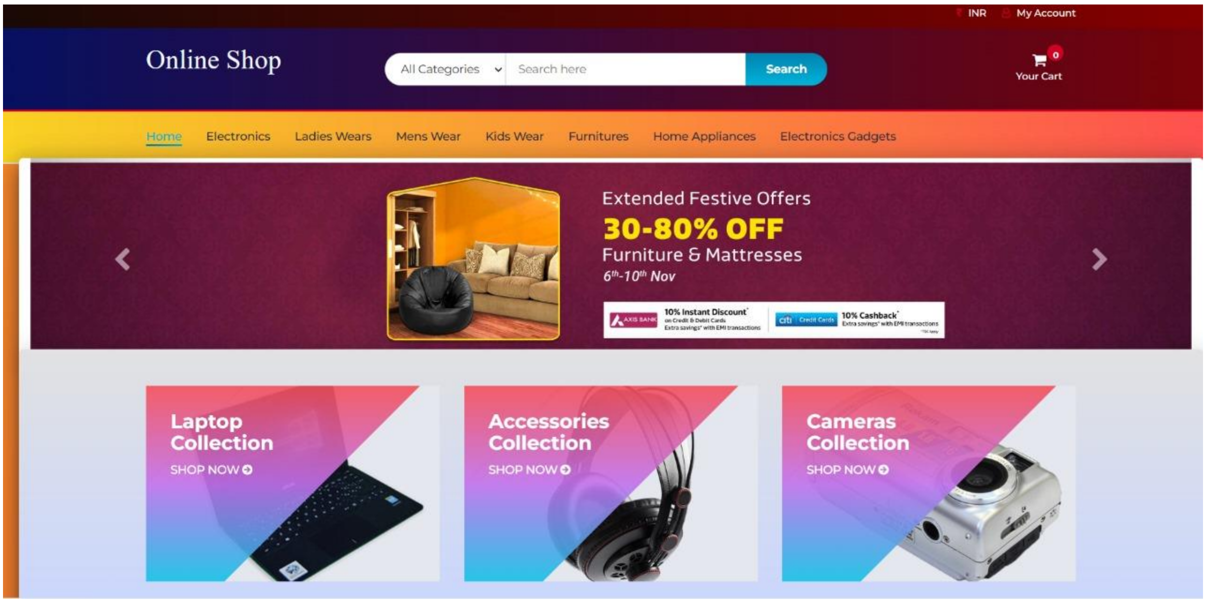


Figure 7.1 Home page

7.2 Categories

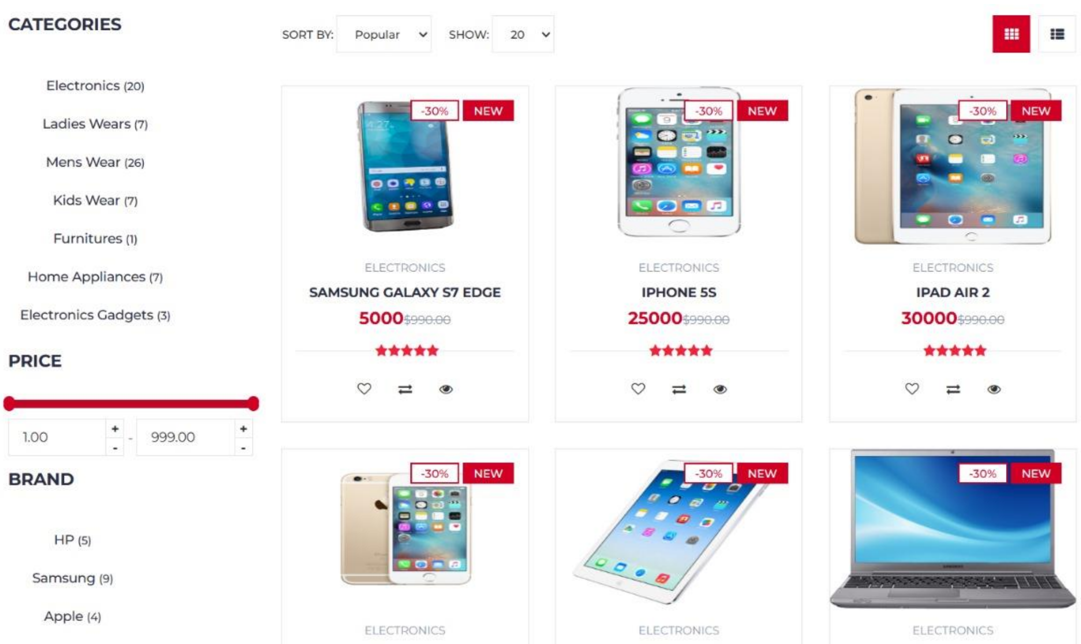


Figure 7.2 Categories

7.3 Dashboard

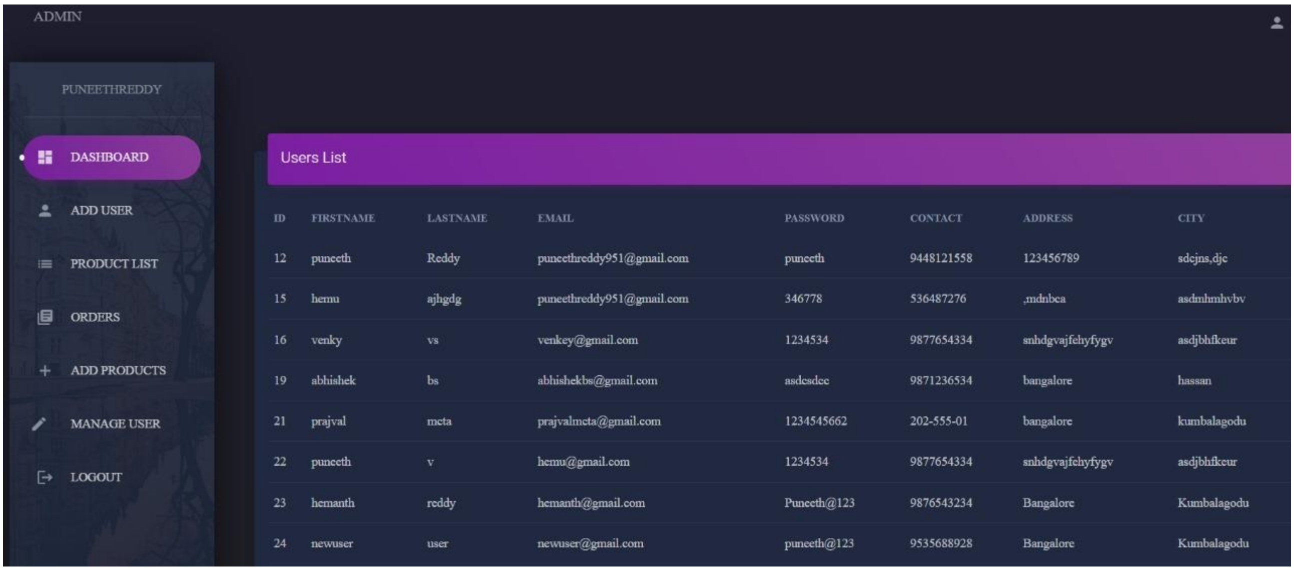


Figure 7.3 Dashboard

7.4 Answered Enquiries

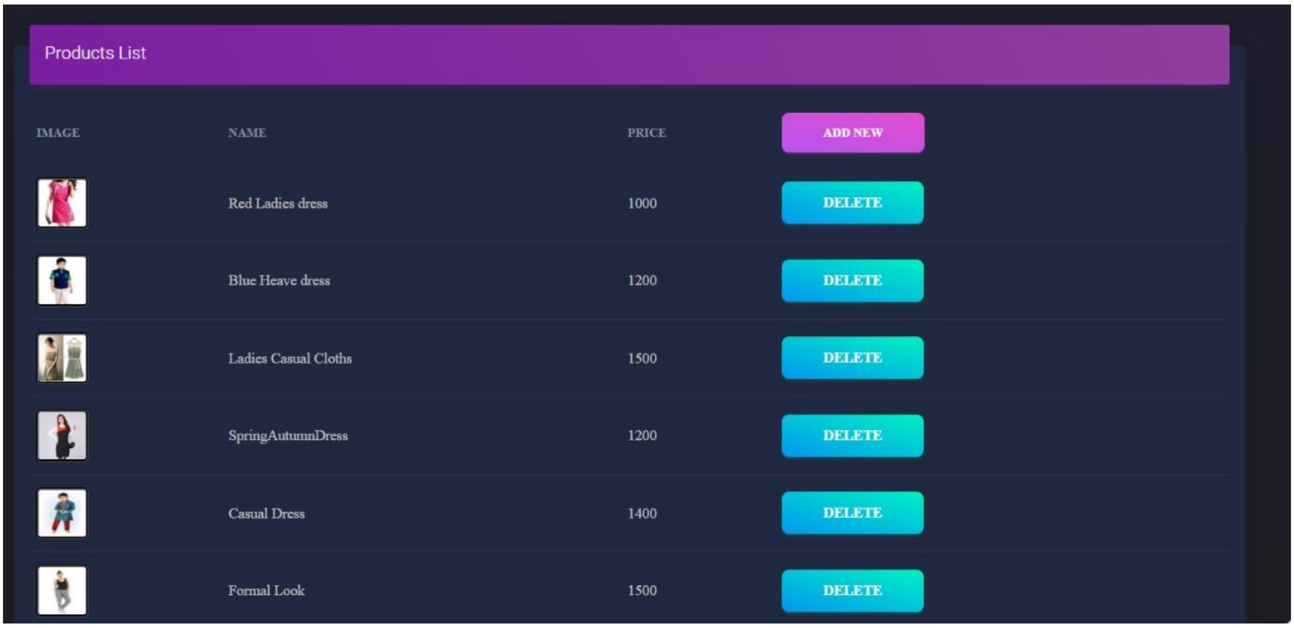


Figure 7.4 Answered Enquiries

7.5 Order page

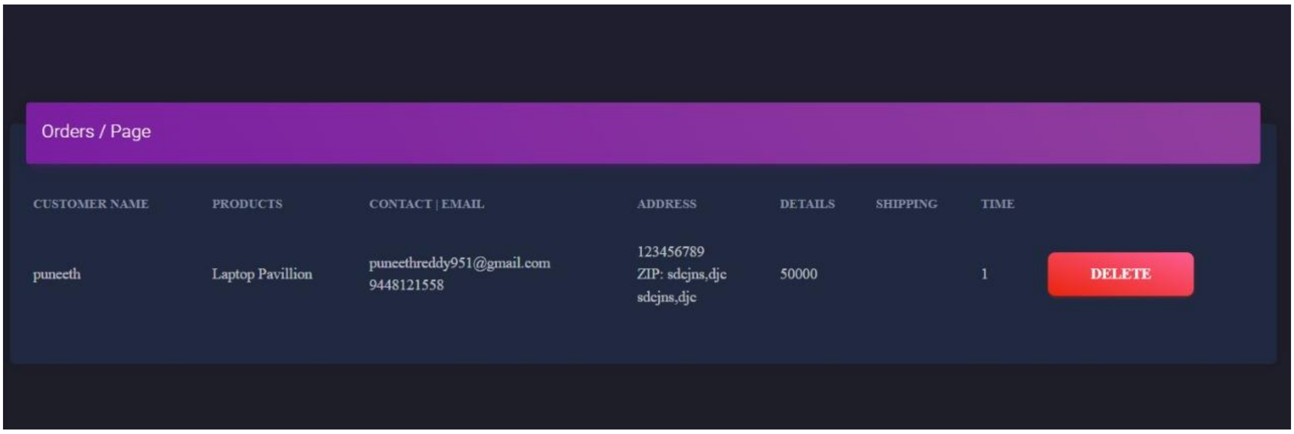


Figure 7.5 Order Page

7.6 E-Commerce System database page

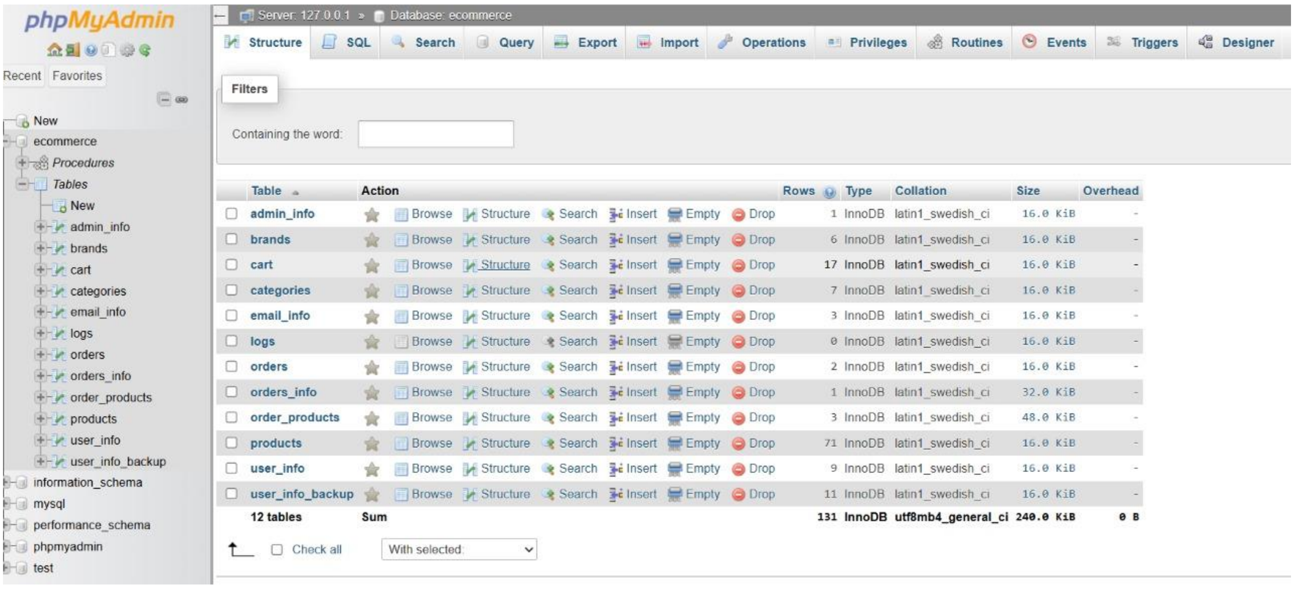


Figure 7.6 E-Commerce Management database page

Module 8:

CONCLUSION

The primary focus of this project was on implementing a robust **Database Management System (DBMS)** that forms the backbone of the entire application. By using a relational database model, we efficiently handled multiple entities like users, products, orders, and payments, ensuring that data is organized, consistent, and easily accessible. The normalization of the database schema allowed us to eliminate data redundancy and achieve data integrity, which is essential for the proper functioning of the system, especially when scaling for larger numbers of products and users..

In terms of performance, the system has been designed to handle an increasing amount of data as more users and products are added to the platform. We implemented optimized **SQL queries** to ensure fast data retrieval, and by indexing frequently accessed tables, we enhanced the system's performance, even as the volume of transactions and products grows. The system can easily scale to accommodate additional features such as product reviews, ratings, recommendations, and advanced analytics in the future.

The **Online Shopping System** project has thus provided an excellent learning opportunity in terms of system design, database management, and web application development. The successful implementation of this system proves that, with proper planning, a carefully designed database, and attention to user experience, it is possible to build a fully functional and scalable e-commerce platform. This project serves as a stepping stone toward building more complex and feature-rich online stores in the future, capable of handling greater user demands and offering a wider range of services.

In conclusion, this mini-project not only highlights the technical aspects of developing an online shopping system but also reflects the core principles of modern e-commerce systems, where user experience, data management, and security play crucial roles. The knowledge gained through this project lays the foundation for future developments in web-based shopping systems and serves as an ideal model for real-world applications in the growing field of e-commerce.

Module 9:

FUTURE ENHANCEMENT

Limitations of the System:

1. This project does not give the information about the stock (quantity) present within the shop.
2. This project does not create monthly, yearly Reports.

These are the limitations which further has to be enhanced.

Enhanced artificial intelligence (AI) and machine learning capabilities are also set to revolutionize shopping websites. AI algorithms can analyze vast amounts of data to personalize recommendations, predict user preferences, and optimize products . This not only improves the efficiency of inventory management but also enhances the overall user experience by offering suggestions and promotions based on individual behavior patterns.

Moreover, blockchain technology holds promise for enhancing transparency and security in transactions and data management within the product industry. By providing immutable records of product provenance, sales histories, and customer reviews, blockchain can reduce disputes, enhance trust, and streamline processes across the art ecosystem.

As these innovations continue to unfold, E-Commerce management website systems are poised to become more than just shopping platforms; they will evolve into comprehensive cultural hubs that prioritize sustainability, personalization, efficiency, and seamless user experiences. By embracing these advancements, the future of online shopping websites looks brighter than ever, promising to redefine how people access and experience art worldwide.

Module 10:

REFERENCES

This document contains provisions which, through reference in this text, constitute provisions of the present document.

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